

surface of the rocket varies.

3a

$$\begin{aligned}\text{period} &= 0.36 \text{ s} \\ \text{acceleration} &= \omega^2 x_0 \\ &= (2\pi/0.36)^2 \times 1.7 \times 10^{-2} \\ &= 5.2 \text{ m s}^{-2}\end{aligned}$$

3b

$$\begin{aligned}\text{either kinetic energy} &= \frac{1}{2} m \omega^2 (x_0^2 - x^2) \\ \text{or potential energy} &= \frac{1}{2} m \omega^2 x^2 \quad \text{and} \quad \text{potential energy} = \text{kinetic energy} \\ \frac{1}{2} m \omega^2 (x_0^2 - x^2) &= \frac{1}{2} \times \frac{1}{2} m \omega^2 x_0^2 \quad \text{or} \quad \frac{1}{2} m \omega^2 x^2 = \frac{1}{2} \times \frac{1}{2} m \omega^2 x_0^2 \\ x_0^2 &= 2x^2 \\ x &= x_0 / \sqrt{2} = 1.7 / \sqrt{2} \\ &= 1.2 \text{ cm}\end{aligned}$$

3c

